

Readable, Usable, Steerable: Shared Cross-Lingual Concept Geometry and a Geometry-Aware Steering Primitive in a Multilingual LLM

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Abstract

We ask how a multilingual large language model represents two ordinal concepts—*days of the week* (a cyclic concept) and *years* (a linear one)—across English (EN) and Chinese (ZH), and whether the *geometry* of those representations can be read out and causally used across languages. Working in QWEN2.5-7B-INSTRUCT with a 1.7B cross-scale companion, and gating every claim on random-equal-norm and unrelated-concept controls, we report three complementary findings. **Readable:** a probe trained on activations in one language recovers the concept in the other *without* re-centering (years $r \approx 0.99$; days $r = 0.66\text{--}0.85$ at 7B, up from ~ 0.1 at 1.7B—the cross-lingual co-location *strengthens with scale*), and the day representations carry order-specific ring topology in both languages (order-aware permutation $p \approx 0.003$; a relational control fails). **A methodological negative we treat as a contribution:** cross-lingual representational-similarity (RSA) *magnitude* saturates—unrelated nouns align at $r \approx 0.47$ and days $\approx$ kinship—so magnitude-based similarity is the wrong instrument for “shared manifold” claims. **Usable:** a steering vector built in one language causally moves the model’s prediction in the other, cleanly for the linear concept (years, both directions, beating both controls at the within-language ceiling) but only fragilely for the circular one (days steers, then destroys fluency outside a narrow window). **Steerable:** a norm-preserving, on-manifold (spherical Slerp) intervention is a *cleaner* steering primitive than naive additive steering—most strikingly for days ZH→EN (clean ceiling 2.24 vs. 0.58)—but the benefit is direction-dependent (days EN→ZH fails) and *not* uniquely circular (years also improves), established under an explicit power check. All results are preliminary and pre-registered; we report what survives strict controls and flag what does not.

1 Introduction

A multilingual language model that answers a question correctly in English and in Chinese might be doing one of two very different things. It could be routing each language through a *shared* internal representation of the underlying concept, or it could be maintaining two separate machineries that happen to agree at the output. The difference matters for interpretability, for cross-lingual transfer, and for safety: if concepts are represented in a language-agnostic geometry, then an intervention validated in one language might be expected to act in another. A growing line of work argues that large models converge toward shared, even modality-agnostic, representational spaces (Huh et al., 2024), and that multilingual transformers operate in a latent space with an English-leaning “pivot” (Wendler et al., 2024). What is less settled is whether that sharing holds at the level of concrete *concept geometry*—the shape a small family of related values traces out in activation space—and whether that geometry is not just *readable* but *causally usable* across languages.

We study this with two deliberately chosen concepts whose geometry is known to differ. Days of the week form a *cycle*: Monday follows Sunday, so the seven values are expected to lie on a closed ring, a structure recently documented as a “clock”-like circular feature in language models (Engels et al., 2024). Years form a *line*: they are ordered but not cyclic. This linear-versus-circular contrast is not incidental—it is a lever. A naive additive steering vector that pushes “value $\rightarrow$ value + 1” moves along a line cleanly, but cuts *through* a ring as a

Status. Every result in this paper is *preliminary* (our pre-registration flag `met=false`): the experiments are controlled and pre-registered, but no claim is ratified as confirmatory. We are explicit throughout about which findings clear the strict pre-registered bars and which do not (§8, Appendix B).

chord rather than following its arc, so geometry should predict when a simple intervention suffices and when it does not.

We approach the question through three lenses, each a separate experiment with its own pre-registered controls.

Readable (H1). Can a probe trained on one language’s activations read the concept off the other language’s activations, and do the two languages trace the same shape? We measure this with trained cross-lingual decoders and with a battery of representational-similarity metrics.

Usable (H2). Does a steering vector *built* in one language causally move the model’s behavior in the *other* along the concept—and does concept geometry predict how that steering behaves?

Steerable (H3). When naive additive steering struggles with the circular concept, does a geometry-aware, norm-preserving intervention that rotates the activation *along* the manifold (You et al., 2026) do better?

Contributions.

1. Cross-lingual concept *sharing* is decode-confirmed for both days and years and *strengthens with scale* (a small convergence signal measured by decoding accuracy versus model size, not by similarity magnitude versus size), consistent with the Platonic Representation Hypothesis (Huh et al., 2024).
2. A *methodological negative*: cross-lingual RSA *magnitude* saturates—an unstructured noun floor aligns at $r \approx 0.47$ at both scales we test, and the circular concept’s similarity is statistically indistinguishable from a relational control. The field’s default similarity instrument cannot support shared-manifold *magnitude* claims; the survivors are *RSA-independent* (a trained decoder and an order-aware topology test).
3. *Causal usability* dissociates by concept geometry: the linear concept (years) steers cleanly cross-lingually; the circular one (days) is steerable but *fragile*—naive additive steering moves it strongly while *destroying fluency*, which is degradation, not mere resistance.
4. A *geometry-aware steering primitive* (norm-preserving spherical rotation) is *cleaner* than additive steering—a result we qualify carefully

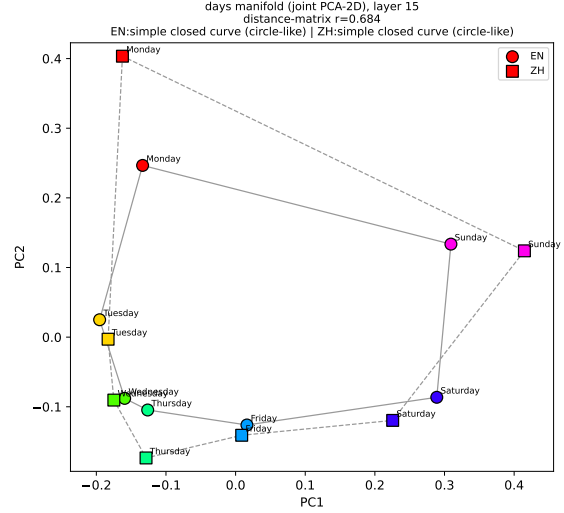


Figure 1: **The shared day ring.** Top-2 principal components of the day-of-week activations in QWEN2.5-7B-INSTRUCT, English (circles) and Chinese (squares), with the semantic-order path overlaid. Both languages trace a simple closed curve through the seven days; the two rings are similarly placed (low Procrustes disparity) but the cross-lingual distance-matrix correlation here is best read as a shape descriptor rather than a magnitude claim (§4).

as direction-dependent and not uniquely circular, and establish under a power check that the two interventions are genuinely distinct.

Figure 1 shows the headline geometric object: the shared EN/ZH day ring. We emphasize at the outset that our strongest positive evidence is RSA-independent, that the RSA-saturation result is a contribution rather than a caveat, and that the H3 “win” is genuinely qualified. We never claim “geometry dictates the intervention” or that on-manifold steering “solves” the circular concept.

2 Related Work

Activation steering. Adding a contrast vector to the residual stream to steer generation was popularized as Activation Addition (ActAdd) (Turner et al., 2023) and sits within a broader representation-engineering program (Zou et al., 2023), with inference-time interventions shown to elicit specific behaviors such as truthfulness (Li et al., 2023). ActAdd is our additive baseline. *Geometry-aware* steering—rotating the activation along the unit sphere toward a target direction while preserving its norm—was recently proposed by You et al. (2026); we adopt their spherical (Slerp) intervention as our on-manifold method, noting two docu-

mented deviations (we omit their task-specific confidence gate, which is specific to truthfulness rather than concept-successor steering, and we anchor at a fixed layer; see §3).

Concept geometry. Language models encode some ordered concepts as low-dimensional manifolds, including linear representations of space and time (Gurnee and Tegmark, 2024) and explicitly *circular* features for cyclic concepts such as days and months (Engels et al., 2024). Our linear/circular contrast is chosen precisely to exercise this distinction.

Cross-lingual representations. Multilingual encoders and decoders are widely held to learn a partially language-neutral space (Conneau et al., 2020; Libovický et al., 2020; Artetxe and Schwenk, 2019), and recent mechanistic work describes an English-pivot effect in the latent language of multilingual transformers (Wendler et al., 2024). The Platonic Representation Hypothesis (Huh et al., 2024) predicts that such sharing should *increase* with scale; we report a decode-versus-scale signal consistent with it.

Similarity tooling—and its limits. We use the standard toolkit: representational similarity analysis (RSA) (Kriegeskorte et al., 2008), centered kernel alignment (CKA) (Kornblith et al., 2019), SVCCA (Raghu et al., 2017), and Procrustes analysis. A central methodological message of this paper is that the *magnitude* of cross-lingual RSA saturates and therefore cannot, by itself, license shared-manifold claims—an instrument caveat we make quantitative in §4.

3 Setup and Shared Methods

Model and numerics. Our headline model is QWEN2.5-7B-INSTRUCT (Qwen Team, 2024), run through TransformerLens (Nanda and Bloom, 2022); a cross-scale companion, QWEN3-1.7B-BASE, is run through nnsight (Fiotto-Kaufman et al., 2024). A single backend-agnostic model interface hides the two libraries. Weights are loaded in bf16, but every cached activation and every logit is cast to fp32 before any computation, because the model’s language-model head is bf16 and small differences vanish otherwise.

Concepts and controls. We study *days* (7 values, expected geometry: circle) and *years* (a contiguous range, expected geometry: line), plus two controls:

kinship (8 relational terms, a *structure-specificity* control) and *nouns* (10 arbitrary concrete nouns, a *floor*). English and Chinese values are matched by an explicit ordinal index (Monday ↔ 星期一 because both are index 0); the alignment lives in the data, not in code. Where Chinese lexicalizes a distinction that English marks with a modifier (older vs. younger brother ↔ 哥哥 vs. 弟弟), kinship pairs are formed honestly; one-to-many terms are *excluded*, never silently mis-paired.

Extraction position and scoring. We read the residual stream at the *readout token*—the first token immediately *after* the concept span—because at that point the model has finished composing the concept and must hold it to predict what comes next, the most likely site of a language-agnostic representation before vocabulary routing. This choice also avoids a Chinese-specific trap: the word for Wednesday, 星期三 (*xīngqī sān*, literally “week three”), is tokenized as a shared prefix 星期 plus 三 (“three”), so reading *inside* the span would capture “three”, not “Wednesday”. For the same reason, multi-token targets are scored by the *sum of token log-probabilities* (teacher-forced): all seven Chinese day words share their first token 星期 and differ only later, so first-token scoring cannot separate them.

Controls philosophy. Cross-lingual similarity is pervasive (§4), so an uncontrolled effect is not a concept effect. Every representational or causal claim is gated against a *random-equal-norm* control (a random vector matched to the real vector’s norm) and, where relevant, an *unrelated-concept* control (a kinship vector, norm-matched). *The informative quantity is the gap to controls, not the absolute magnitude.* For steering we additionally require *off-target preservation*: an effect that is really the model degrading into nonsense is not a success, so we measure whether non-concept tokens survive (KL divergence and top-1 agreement).

A one-line glossary. For readers outside interpretability: *RSA* correlates the within-language pairwise-distance matrices of the two languages (same *shape*?); *CKA* measures feature-space similarity invariant to rotation and scale; *Procrustes disparity* is the residual after optimally rotating/scaling one cloud onto the other (low = same *placement*); a *cross-lingual decoder* is a probe trained on one language and tested on the other (can the concept be *read* across languages?); and

the *order-aware ring test* asks whether the *semantic* order of the values traces an unusually short closed tour relative to random orderings (is the cyclic structure really there?).

Reproducibility and pre-registration. Every run writes a manifest (source hash, config hash, model revision, dtype), and the activation cache is keyed by `run_id = <code-hash>_<config-hash>` so stale activations can never feed changed code. Endpoints and thresholds were fixed before the confirmatory runs; amendments are dated and reasoned (notably a cosine $\rightarrow$ Euclidean distance fix and a reframing of H2), and are reproduced in Appendix B.

4 H1: Shared Cross-Lingual Concept Geometry

Headline (honest): on the strict pre-registered bars, H1 does *not* confirm—cross-lingual RSA magnitude saturates. The defensible positives are *RSA-independent*: cross-lingual decoding shows both concepts co-located, and an order-aware test shows days-specific ring topology.

4.1 Decode co-location, and it strengthens with scale

A ridge probe trained to predict the concept value from one language’s activations and tested on the other’s recovers years almost perfectly in both directions, and—crucially—does so *without re-centering* the two languages (Table 1). For days, the *uncentered* decoder reaches $r = 0.66\text{--}0.85$ at 7B. This is the strongest positive in the paper because a trained decoder does not use the saturated similarity metric, and because the days result *strengthens with scale*: the same uncentered decoder reaches only $r \approx 0.10\text{--}0.15$ in the 1.7B companion (Table 4). Following Huh et al. (2024), we read the 1.7B $\rightarrow$ 7B strengthening as a small convergence signal—measured by *decoding* versus scale, not by similarity magnitude versus scale.

4.2 Days-specific ring topology

RSA magnitude is saturated (next subsection), and bare ring-closure is “order-privileged” (any closed tour through points can look like a ring). The discriminating test asks instead whether the *correct* semantic order is special: does Monday $\rightarrow$ Sunday $\rightarrow$ Monday trace an unusually short closed tour relative to thousands of random orderings? It does, in *both* languages, for days

concept	direction	uncentered r	centered r
days	EN $\rightarrow$ ZH	0.657	0.965
days	ZH $\rightarrow$ EN	0.852	0.940
years	EN $\rightarrow$ ZH	0.995	0.995
years	ZH $\rightarrow$ EN	0.995	0.995

Table 1: **Cross-lingual decoding** (7B), RSA-independent. Years are co-located *uncentered*; days decode well uncentered too, with a modest centering gain (a small language offset). A probe trained in one language reads the other.

concept	RSA r	ring p (EN/ZH)	role
years	0.978	0.0002 / 0.0002	positive (linear)
days	0.853	0.0028 / 0.0028	target (circular)
kinship	0.856	0.056 / 0.055	structure control
nouns	0.466	0.46 / 0.94	floor (gate)

Table 2: **The table that tempers the headline** (7B). RSA *magnitude* fails to separate days (0.853) from kinship (0.856), and the noun floor sits at 0.466 (the pre-registered saturation gate required < 0.30). The order-aware ring test *does* separate days from kinship, and is RSA-independent.

($p \approx 0.003$), while the relational kinship control does *not* ($p \approx 0.055$); see Table 2. Because this test never touches the saturated magnitude metric, it supports a defensible *days-specific* topological claim (at $p < 0.05$, not the strict $p < 0.001$). Intrinsic dimensionality separates the structured concepts from the noun floor but does *not* separate days from kinship—so the order-aware test, not dimensionality or RSA magnitude, is the discriminator. Figure 2 shows that this shared shape is present in a broad band of layers, not a single sharp peak, and sits well above the shuffled null while Procrustes disparity sits well below its null (“same shape, similarly placed”).

4.3 The methodological negative: RSA magnitude saturates

Our pre-registered floor control (arbitrary nouns) was required to fall below an RSA of 0.30. It does not: unrelated nouns align at $r = 0.466$ at 7B and 0.46 at 1.7B—essentially flat across roughly $4\times$ scale, so the saturation is robust rather than a small-model artifact (Table 4). Worse for any magnitude claim, the circular concept’s RSA (0.853) is statistically indistinguishable from the relational kinship control (0.856; gap -0.003). Per our pre-committed interpretation grid (Appendix B), this means *no RSA-magnitude geometry claim stands*: cross-lingual representational similarity is perva-

metric	days	years
<i>shape-intrinsic (same shape?)</i>		
RSA (distance-matrix r)	0.853	0.978
linear CKA	0.948	0.987
RBF CKA	0.963	0.988
intrinsic dim. (EN/ZH)	3.97/3.32	3.16/4.26
<i>embedding-dependent (same place?)</i>		
Procrustes disparity	0.042	0.033
principal-angle cos	0.397	0.714

Table 3: **Two metric families at the report layer (7B).** Shape is high and placement is close for both concepts; under the failed saturation gate the magnitude rows are exploratory, not gated positives. Procrustes disparities are far below their shuffled nulls (0.29 for days, 0.90 for years).

sive even for unstructured vocabulary, so RSA magnitude is the wrong instrument for “shared manifold” conclusions. We treat this as a contribution: it explains why our positive evidence is built on the decoder and the topology test rather than on similarity magnitude. Table 3 reports the full two-family battery for completeness, with the explicit understanding that the magnitude rows are exploratory under the failed gate.

4.4 The pre-registered RSA endpoint, reported honestly

Our single confirmatory endpoint was the days RSA at a layer selected on a held-out split. After a pre-registered fix—raw cosine distance is not invariant to a point-cloud shift, and we found a cross-lingual offset—the *canonical* distance is Euclidean. On the held-out split the canonical Euclidean RSA is 0.684 ($p = 0.0055$): it clears the effect-size bar (≥ 0.50) but *misses* the significance bar ($p < 0.001$). The translation-invariant centered-cosine variant reaches 0.865 ($p = 0.0005$) and would clear both, so the endpoint is distance-dependent. Either way, under the failed floor gate the magnitude reading is exploratory. We therefore report H1 as *partial / reframed*, leading with the RSA-independent positives.

4.5 Behavior and geometry dissociate

The 1.7B base model fails English day-succession behaviorally (it scores $\sim 21\%$ across raw, alternate-phrasing, and chat-template elicitations; the chat-template fix that recovers the *instruct* sibling does not help the base model). Yet that same base model carries a clean, ZH-shared day ring internally. This is *behavioral failure despite representa-*

metric	1.7B-Base	7B-Instruct
noun-floor RSA (gate < 0.30)	0.46	0.466
days RSA — kinship RSA	−0.01	−0.003
days decode (uncentered)	0.10–0.15	0.66–0.85
years decode (uncentered)	~ 0.997	~ 0.995
days ring p (EN/ZH)	0.003/0.003	0.003/0.003
kinship ring p (EN/ZH)	0.06/0.16	0.056/0.055
offset cos / transfer	0.64/0.50	0.45/0.32

Table 4: **Cross-scale robustness** (1.7B-Base vs. 7B-Instruct). The saturation floor is flat across scale (the negative is robust); days decode *strengthens* with scale (a convergence signal); the ring topology replicates; and the offset-axis finding does *not* replicate (retired). No run larger than 7B is claimed.

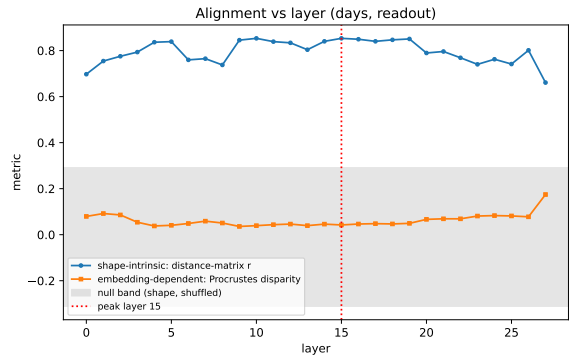


Figure 2: **Shape vs. placement across layers** (days, 7B, readout position). Shape-intrinsic similarity (blue) is high in a broad band and well above the shuffled null (grey); embedding-dependent Procrustes disparity (orange) is low. The dotted line marks the held-out-selected layer. The shared shape is not a single sharp mid-layer spike.

tional structure—elicited next-token behavior and internal geometry come apart—and is exactly why we measure H1 on geometry rather than on a behavioral quiz; the geometry reproduces on the instruct model.

One finding we retire. On the 1.7B model we observed a single “language offset” direction shared between the two concepts. At 7B this does not replicate (the offset cosine drops to 0.45 and cross-transfer to 0.32; the conclusion flips to per-concept centering), plausibly because days are *more* co-located at 7B and so share a smaller offset. We retire this finding rather than carry it forward (Table 4).

5 H2: Causal Cross-Lingual Steering

H1 shows the geometry is *readable*; H2 asks whether it is *causally usable*. We build a

dir	coef	eff	vs. rand	vs. unrel	$\approx$ within
EN→ZH	8	0.520	beats all	0.52/0.24	0.44
ZH→EN	12	0.482	beats all	0.48/0.14	0.49

Table 5: **Years steer cleanly** (7B, layer 19). Off-target preserved at the claim coefficient; effect is concept-specific (beats the unrelated control) and at the within-language ceiling.

steering vector from the per-pair contrast “value $\rightarrow$ value +1” in the *source* language and inject it at the prompt positions of a neutral, concept-eliciting prompt in the *target* language (e.g. ZH 我最喜欢的一天是, “my favorite day is”), scoring whether the model shifts toward the successor. The effect is the steered-minus-unsteered change in log-probability of the successor relative to the current value, averaged over consecutive pairs and prompts. A methodological note: a single *mean* “+1” vector telescopes to zero around a closed loop (consecutive differences around a ring sum to zero), so we average the *effect* of per-pair vectors, never the vectors.

5.1 Years: clean cross-lingual steering

For the linear concept, steering transfers cleanly in both directions (Table 5). The effect beats all random-norm draws and the unrelated-concept control, sits at the within-language ceiling, preserves off-target tokens, and holds across a broad coefficient range. EN→ZH and ZH→EN are comparable, so there is no strong English-pivot asymmetry *for steering*; the result reproduces across independent reruns. This is a causal complement to the H1 decode finding: cross-lingual concept representations are not just *readable* across languages, they are *causally usable*.

5.2 Days: steerable but fragile

The circular concept *does* steer cross-lingually and concept-specifically, but only in a narrow low-coefficient window; above it, naive additive steering produces a large effect while *destroying* fluency (Table 6, Figure 3). This is the crucial honesty distinction: a concept that “fails” a steering gate can fail in two opposite ways—*no effect* (genuine resistance) or *a large effect that breaks off-target* (destruction). Days is the latter. The linear concept tolerates roughly 4–8 $\times$ more steering strength before off-target degrades. We call this a *coefficient-sensitivity* dissociation, consistent with circular ge-

dir	clean win.	clean eff	high-coef behavior
EN→ZH	$c = 0.5$	0.102	$c=4$: 0.66, off-tgt <i>broken</i>
ZH→EN	$c = 1-2$	0.578	$c=4$: 1.07, off-tgt <i>broken</i>

Table 6: **Days are fragile** (7B, layer 19). Clean only in a narrow low-coefficient window; above it the effect grows but fluency collapses (degradation, not resistance).

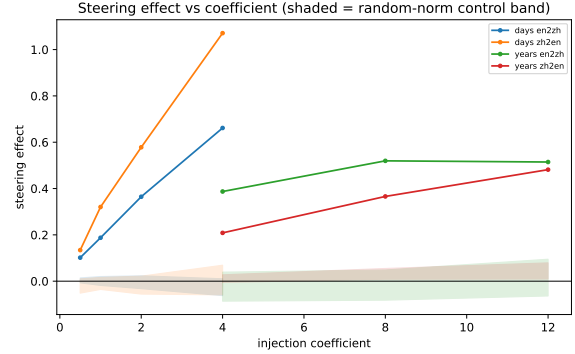


Figure 3: **Steering effect vs. coefficient** (7B). Days (blue/orange) respond strongly at small coefficients but their clean window is narrow; years (green/red) rise gradually and tolerate much larger coefficients. Shaded bands are the random-norm control.

ometry being harder to steer without falling off-manifold.

5.3 The geodesic test here was underpowered

We attempted an on-manifold (geodesic) contrast within H2 but it did not beat the naive chord—and *was underpowered*: at adjacent-day granularity the naive chord is already a secant between real activations, nearly on-manifold, and a synthetic two-dimensional tangent is not clearly *more* on-manifold. The chord and the tangent are nearly the same step, so this test neither supports nor refutes geometry-aware steering. It motivates a properly powered test (H3); H2 does *not* predict an H3 win.

6 H3: Geometry-Aware (Norm-Preserving) Steering

Headline (qualified): norm-preserving spherical steering is a *cleaner* cross-lingual steering primitive than naive additive steering—most dramatically for days ZH→EN—but the evidence is direction-dependent (days EN→ZH fails) and not uniquely circular (years also benefits). We do *not* claim that geometry dictates the intervention, nor that on-manifold steering solves the circular concept.

concept	dir	add.	sph.	cleaner?
days	ZH→EN	0.578	2.236	yes (3.9×)
days	EN→ZH	0.102	0.000	no—fails
years	EN→ZH	0.520	0.656	yes (1.26×)
years	ZH→EN	0.482	1.158	yes (2.40×)

Table 7: **Per-direction clean ceilings** (7B, layer 19): additive vs. spherical. The spherical norm ratio is 1.00 in every cell (power check); additive is 1.05–1.32.

The two interventions. Both use the same per-pair contrast *direction* $\mu = \text{unit}(h_{v+1} - h_v)$. Naive additive steering is $h' = h + c\mu$, which changes the activation norm. The on-manifold method rotates the unit activation toward μ by a Slerp fraction t and rescales to the original norm:

$$\hat{h}' = \frac{\sin((1-t)\theta)}{\sin\theta} \hat{h} + \frac{\sin(t\theta)}{\sin\theta} \mu, \quad h' = \|h\| \hat{h}', \quad (1)$$

where θ is the angle between $\hat{h}$ and μ . This is norm-preserving by construction, the property we hypothesize avoids the off-manifold blow-up that destroys fluency.

The clean ceiling and the power check. For each method we report the *clean ceiling*: the largest steering effect achieved while *both* beating the random-norm control and preserving off-target. A method “helps” only if its clean ceiling is materially higher. Because the two methods must be geometrically distinct for the comparison to mean anything, we verify a *power check*: spherical preserves the norm (ratio 1.00 everywhere) while additive grows it (1.05–1.32). This is the power the H2 geodesic test lacked, so a null here would be a real null.

6.1 Results

Table 7 reports per-direction clean ceilings, and Figure 4 is the money figure—steering effect versus off-target degradation, with on-manifold (blue) dominating naive (red) where it works and breaking off-target where it does not.

Days ZH→EN—the win. Spherical rotation reaches a clean ceiling of 2.236 with off-target preserved across the entire sweep, and is concept-specific (the norm-matched unrelated vector gives -0.296 , i.e. it moves days, not a generic direction). Naive additive caps at a clean 0.578 and breaks off-target above it. Where the naive chord must choose between a small clean effect and a large destructive one, norm-preserving rotation delivers a large

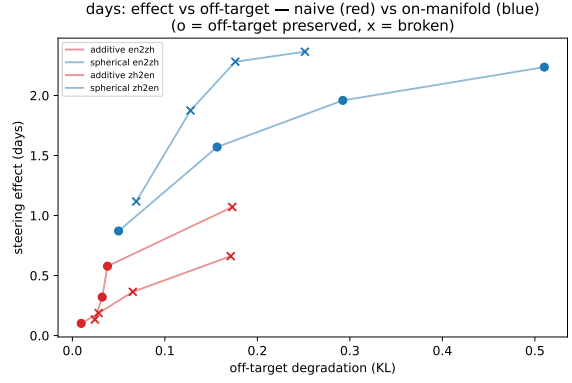


Figure 4: **The money figure** (days, 7B). Steering effect vs. off-target degradation (KL), naive additive (red) vs. on-manifold spherical (blue); filled markers preserve off-target, crosses break it. The ZH→EN spherical curve climbs to a large effect while staying clean; the EN→ZH spherical curve breaks off-target at every step.

effect that stays clean—the predicted mechanism.

Days EN→ZH—the failure. In the other direction spherical *does not* work: the clean ceiling is 0.000. It produces large raw effects (up to 2.365) but off-target is broken at *every* Slerp fraction. Norm preservation does not rescue this direction; on-manifold steering does *not* solve days globally. This asymmetry is the single most important qualifier on the result, and it travels with the headline.

Years also benefits—a replication, and the catch. The linear anchor also gets cleaner under spherical steering in both directions (0.656 vs. 0.520; 1.158 vs. 0.482). This is a useful positive control—a faithful cross-lingual replication of [You et al. \(2026\)](#)—but it directly weakens the claim that the benefit is uniquely due to *circular* geometry, because a concept with no ring structure improves too. The largest single gain is on the circular concept (days ZH→EN, 3.9×) and exceeds the years EN→ZH gain (1.26×), so the effect *leans* circular; but years ZH→EN gains a comparable 2.40×, so circular-specificity is partial, not established. The honest reading is “norm preservation is a generally cleaner steering primitive”, not “circular geometry dictates the intervention”.

7 Discussion

One arc, three legs. Cross-lingual concept geometry is *readable* (a decoder trained in one language reads the other), *causally usable* (a years vector built in English moves the model’s Chinese prediction and vice versa), and—for one concept and

one direction—*cleaner to steer on-manifold* (days ZH→EN). Each leg is controlled, and each is preliminary.

Two negatives that carry weight. First, cross-lingual RSA magnitude saturates: a default similarity instrument reports high agreement even for unstructured nouns, so magnitude alone cannot license shared-manifold claims. This is a tooling lesson with consequences beyond our concepts. Second, the H3 benefit is direction-dependent and not uniquely circular—a scoped result, not a sweeping one. We think reporting both negatives plainly is part of the contribution.

Scale. The convergence signal lives in *decoding* versus scale (days co-location rose from ~ 0.1 to $0.66\text{--}0.85$ from 1.7B to 7B), consistent with the Platonic Representation Hypothesis. The RSA floor does *not* un-saturate with scale, so we deliberately report decode-versus-scale, not RSA-versus-scale, and claim no model larger than 7B.

Why on-manifold steering helps when it helps. Preserving the activation norm avoids the off-manifold blow-up that destroys fluency. But it helps *generally*, including for the linear concept, so the right framing is “norm preservation is a cleaner primitive” rather than “circular geometry requires it”. Why the benefit reverses for days EN→ZH is open; candidate causes include the multi-token Chinese target tokenization, anisotropy between the EN and ZH day sub-manifolds, and layer specificity. We report the asymmetry as a finding, not an explanation.

8 Limitations

This paper is preliminary by design (`met=false`); nothing is ratified as confirmatory. (i) *Small value sets* (7 days; an 11-year window for steering) make RSA over $\sim 21\text{--}28$ distance pairs noisy. (ii) *RSA magnitude saturates* and does not separate days from a relational control; we therefore do not rest any claim on it. (iii) *Coarse permutation p* for steering (15 random draws $\rightarrow$ finest $p \approx 0.06$): we report “beats all draws”, not a precise p , and would raise the count for a confirmatory claim. (iv) *A single injection/readout layer and a single knob grid*; results are not swept over layers. (v) The within-language ceiling is itself *noisy* (it can break off-target at some knobs), so treat it as indicative. (vi) The *H3 direction asymmetry is unexplained*. (vii) We omit You et al.’s confidence gate and fix

a layer; both are documented deviations that affect direct comparability to their truthfulness results. (viii) Our strongest confirmatory endpoint (the RSA bar) is *not met*; the surviving positives are RSA-independent and reported at $p < 0.05$.

9 Conclusion

Across English and Chinese, a multilingual LLM represents days and years in geometry that is cross-lingually *readable* (decode-confirmed, strengthening with scale) and *causally usable* (clean for the linear concept, fragile for the circular one), and a norm-preserving spherical intervention is a *cleaner*—though direction-dependent and not uniquely circular—steering primitive than naive addition. The most transferable lesson may be the negative one: cross-lingual similarity magnitude saturates, so claims about “shared manifolds” should rest on trained decoders and order-aware topology rather than on similarity scores alone. We release the pre-registration, manifests, and per-condition metrics to support independent checking.

Ethics and Broader Impact

Activation steering is dual-use: the same mechanism that nudges a model toward a benign successor can be used to bias outputs. Our cross-lingual results suggest that interventions validated in one language may transfer to another, which is relevant to multilingual safety auditing—an intervention checked only in English may already act in Chinese, for better or worse. We study small, benign factual concepts and report preliminary results gated on controls; we make no deployment claims.

Reproducibility Statement

All numbers are produced by config-driven runs with fixed seeds; each run writes a manifest (source/config hash, model revision, dtype) and the activation cache is keyed so changed code cannot reuse stale activations. Appendix A lists the run identifiers and the report-only regeneration path; Appendix B reproduces the pre-registration and its dated amendments.

References

Mikel Artetxe and Holger Schwenk. 2019. Massively multilingual sentence embeddings for zero-

- shot cross-lingual transfer and beyond. *Transactions of the Association for Computational Linguistics (TACL)*, 7:597–610.
- Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. 2020. Unsupervised cross-lingual representation learning at scale. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*.
- Joshua Engels, Isaac Liao, Eric J. Michaud, Wes Gurnee, and Max Tegmark. 2024. Not all language model features are linear. In *arXiv preprint arXiv:2405.14860*.
- Jaden Fiotto-Kaufman, Alexander R. Loftus, Eric Todd, Jannik Brinkmann, Caden Juang, Koyena Pal, Can Rager, Aaron Mueller, Samuel Marks, Arnab Sen Sharma, and 1 others. 2024. NNsight and NDIF: Democratizing access to foundation model internals. *arXiv preprint arXiv:2407.14561*.
- Wes Gurnee and Max Tegmark. 2024. Language models represent space and time. *Proceedings of the International Conference on Learning Representations (ICLR)*.
- Minyoung Huh, Brian Cheung, Tongzhou Wang, and Phillip Isola. 2024. The platonic representation hypothesis. *arXiv preprint arXiv:2405.07987*.
- Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. 2019. Similarity of neural network representations revisited. In *Proceedings of the International Conference on Machine Learning (ICML)*, pages 3519–3529.
- Nikolaus Kriegeskorte, Marieke Mur, and Peter A. Bandettini. 2008. Representational similarity analysis – connecting the branches of systems neuroscience. *Frontiers in Systems Neuroscience*, 2:4.
- Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. 2023. Inference-time intervention: Eliciting truthful answers from a language model. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Jindřich Libovický, Rudolf Rosa, and Alexander Fraser. 2020. On the language neutrality of pre-trained multilingual representations. In *Findings of the Association for Computational Linguistics: EMNLP 2020*.
- Neel Nanda and Joseph Bloom. 2022. Transformerlens. <https://github.com/TransformerLensOrg/TransformerLens>.
- Qwen Team. 2024. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*.
- Maithra Raghu, Justin Gilmer, Jason Yosinski, and Jascha Sohl-Dickstein. 2017. SVCCA: Singular vector canonical correlation analysis for deep learning dynamics and interpretability. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Alexander Matt Turner, Lisa Thiergart, David Udell, Gavin Leech, Ulisse Mini, and Monte MacDiarmid. 2023. Activation addition: Steering language models without optimization. In *arXiv preprint arXiv:2308.10248*.
- Chris Wendler, Veniamin Veselovsky, Giovanni Monea, and Robert West. 2024. Do llamas work in English? on the latent language of multilingual transformers. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*.
- You, Deng, and Chen. 2026. Spherical steering: Geometry-aware activation rotation. In *Proceedings of the International Conference on Machine Learning (ICML)*. ArXiv:2602.08169; code at <https://github.com/chili-lab/Spherical-Steering>.
- Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xu Wang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, and 1 others. 2023. Representation engineering: A top-down approach to ai transparency. *arXiv preprint arXiv:2310.01405*.

A Reproducibility Details

Each phase is launched by a batch script (`run_phase{1,2,3}`) with seed 0 and deterministic kernels. The activation cache is keyed by `run_id = <code-hash>_<config-hash>`; manifests record the resolved config, model name and revision, dtype, device, and library versions. Reports and figures regenerate on CPU from a persisted results object without re-running the GPU forward passes (`--regen`). The headline run identifiers are `wt7d5ac380bf` (H1), `wtbeaf70166f` (H2), and `wtee1c69f9d5` (H3); manifests for all runs accompany the release.

B Pre-Registration and Amendments

The pre-registered confirmatory endpoints were: (1) the days RSA at a held-out-selected layer, ≥ 0.50 and beating the label-shuffled null at $p < 0.001$; (1b) an order-aware ring test passing in both languages at $p < 0.001$ with a relational control failing; (1c) a hard *saturation gate* requiring the noun floor RSA < 0.30 before any RSA-magnitude claim; and (2) cross-lingual steering of a linear concept at ≥ 1.0 normalized-logit-difference SD, beating an equal-norm random control.

Two dated amendments, both recorded *before* the corresponding confirmatory run, are material. First (distance fix): raw cosine distance is not invariant to a point-cloud translation, and we found a cross-lingual offset, so the canonical RSA distance

concept	dir	method	knob	eff	off-tgt
days	EN→ZH	add	0.5	0.102	ok
days	EN→ZH	add	1	0.188	broken
days	EN→ZH	add	2	0.365	broken
days	EN→ZH	add	4	0.661	broken
days	EN→ZH	sph	0.05	1.117	broken
days	EN→ZH	sph	0.10	1.875	broken
days	EN→ZH	sph	0.15	2.281	broken
days	EN→ZH	sph	0.20	2.365	broken
days	ZH→EN	add	0.5	0.134	broken
days	ZH→EN	add	1	0.320	ok
days	ZH→EN	add	2	0.578	ok
days	ZH→EN	add	4	1.071	broken
days	ZH→EN	sph	0.05	0.871	ok
days	ZH→EN	sph	0.10	1.571	ok
days	ZH→EN	sph	0.15	1.959	ok
days	ZH→EN	sph	0.20	2.236	ok
years	EN→ZH	add	4	0.387	ok
years	EN→ZH	add	8	0.520	ok
years	EN→ZH	add	12	0.515	broken
years	EN→ZH	sph	0.05	0.402	ok
years	EN→ZH	sph	0.10	0.652	ok
years	EN→ZH	sph	0.15	0.656	ok
years	EN→ZH	sph	0.20	0.607	broken
years	ZH→EN	add	4	0.209	ok
years	ZH→EN	add	8	0.366	ok
years	ZH→EN	add	12	0.482	ok
years	ZH→EN	sph	0.05	0.417	ok
years	ZH→EN	sph	0.10	0.777	ok
years	ZH→EN	sph	0.15	1.020	ok
years	ZH→EN	sph	0.20	1.158	ok

Table 8: **Full H3 sweep** (7B, layer 19). Every row beats the random-norm control; the clean ceiling is the largest effect with off-target “ok”. Days EN→ZH never preserves off-target under spherical (clean ceiling 0); days ZH→EN preserves it throughout.

was changed to Euclidean (or centered cosine); the original cosine endpoint is retained side-by-side for continuity, and the order-aware ring test was added as the topological endpoint, replacing order-privileged bare ring-closure. Second (H2 reframe): the original “days resist steering” framing was replaced by a pre-committed grid distinguishing *resistance* (no effect) from *destruction* (large effect, broken off-target), after first-run data showed days produce large effects while breaking off-target. Under these commitments, H1’s RSA magnitude is reported as exploratory (the floor gate failed) and H3 is reported as a qualified, direction-dependent, not-uniquely-circular result.

C Full Per-Knob Steering Sweep (H3)

Table 8 gives the per-knob effect and off-target status for both methods, both concepts, and both directions, underlying the clean ceilings in Table 7.

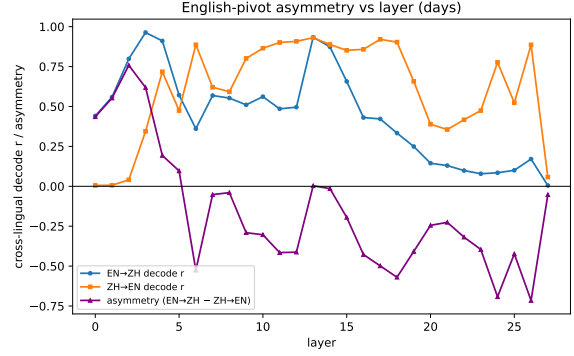


Figure 5: EN→ZH minus ZH→EN decode quality per layer (7B). A positive, English-favoring asymmetry peaks in early layers, consistent with an English-pivot signature (Wendler et al., 2024).

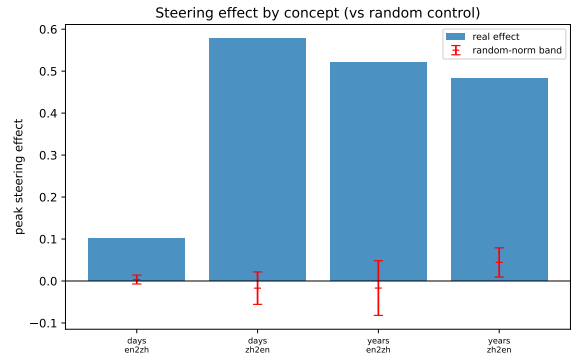


Figure 6: Per-concept cross-lingual steering effect, both directions, with the random-norm control band (7B).

D Supplementary Figures

Figure 5 (English-pivot asymmetry across layers), Figure 6 (per-concept steering effect), and Figure 7 (per-knob effect and off-target for both methods) support the main text. The cross-scale and underpowered-control companions (the 1.7B day ring; the underpowered H2 geodesic) accompany the release.

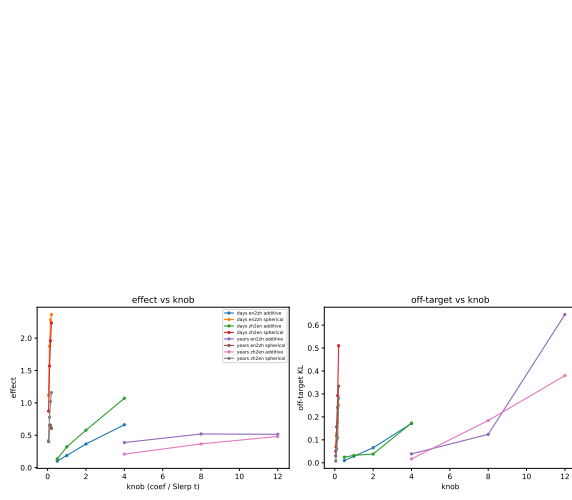


Figure 7: Effect (left) and off-target KL (right) vs. knob for both methods and both concepts (7B). Spherical knobs are small Slerp fractions; additive coefficients span the clean window into destruction.